HSAL-GNN LLM-Based Relation Enhancement

1. Objective

Enhance the existing HSAL-GNN heterogeneous graph recommender by using an LLM's world knowledge, adapted from the GraphGPT idea, to discover narrative relationships between books that are not captured in the training graph and add them as new graph edges.

For each central book, the LLM determines whether a specified relation exists and identifies the related work. The identified work is embedded using the same model as the existing graph, matched to the most similar graph node using cosine similarity, and added as an edge. The effect on HSAL/LRD downstream performance will then be evaluated. **[NOT YET DONE]** Graph integration and retraining have not started.

The project was initially scoped to fully implement and validate **one relation type at a time** rather than all eight categories in parallel: Prequel, Sequel, Side Story, Spin-off, Alternative Version, Midquel, Retelling, and Retcon. It started with **Prequel**, then expanded to **Sequel** using the same structural ground-truth mechanism, and later to **Alternative Version**. The remaining five categories are pending because their definitions have not yet been defined, pending.

2. Dataset

2.1 Goodreads Young Adult

Source: UCSD/McAuley Lab Goodreads dataset, publicly downloadable from https://mcauleylab.ucsd.edu/public_datasets/gdrive/goodreads/. Raw file: goodreads_books_young_adult.json.gz — one JSON object per book, fields include book_id, title, description, authors (list of {author_id, role}), average_rating, ratings_count Interaction file: Goodreads_young_adult_HSAL_100k.csv — the existing HSAL training interactions (user_id, item_id, time, item_series_id), used to determine which book_ids are "active" (i.e., actually appear in the HSAL training data) before extracting metadata for them. Filtering: of the interaction file's unique item_ids, 8,108 were found present in goodreads_books_young_adult.json.gz Author names: goodreads_book_authors.json.gz (17,988,861 bytes, downloaded and confirmed) maps author_id → name; used only for Version B.

3. Dataset Versions: A vs. B

	Version A	Version B
Meta CSV	Goodreads_young_adult_item_meta_with_title.csv	Goodreads_young_adult_author.meta.csv
Meta columns	item_id, title, i_genre_c, r_sequel	item_id, title, description, authors, average_rating, ratings_count, i_genre_c, r_sequel
Embedding text	title + description + genre	title + description + genre + author
Embedding file	Goodreads_young_adult.feat.npy (8108×384)	Goodreads_young_adult_author.feat.npy
Index file	Goodreads_young_adult.item2index	Goodreads_young_adult_author.item2index
Raw text file	Goodreads_young_adult.text	Goodreads_young_adult_author.text
Used in quantitative experiments this session?	Yes – Prequel/Sequel/AltVersion	Yes – full comparison, §13

Why author is included in Version B: The idea was that adding the author could help distinguish books with the same title and give the embeddings more context, Version B is now compared for all three relation types

EXPERIMENTAL_THRESHOLD = 0.5 is the cosine similarity cutoff used to decide whether two book embeddings are similar enough to be considered a match and kept the same 0.5 threshold from Version A for Version B instead of tuning it again

The coherence check suggests the new embeddings are working reasonably well, but we did not separately test whether 0.5 is still the right similarity cutoff for Version B. so, the Version B results use the same cutoff as Version A

Embedding model (both versions): all-MiniLM-L6-v2 via sentence-transformers (installed version 6.0.1 in the local venv), producing 384-dimensional vectors, printed shape (8108, 384) on every run.

Dataset Statistics

- 99,922 interactions → 8,108 distinct items
- **7,665 (94.5%)** are part of a series; **443 (5.5%)** are standalone.
- **4,000 distinct series:** 2,178 with 1 item, 988 with 2, 710 with 3–5, and 124 with 6+.
- **Series positions:** median **1**, mean **2.35**, max **58**.
- Of the 7,665 series items: Has a later entry in same series present in this file (has 'sequel'): 3,665
- No later entry present (either last book, or only book seen): 4,000

6. Relation Type: PREQUEL

6.1 Definition

A Prequel is a book that takes place before the central book and is part of the same series or universe.

6.2 Results — Version A, n=400 (165 with_series / 165 without_series/ 70 ambiguous, informational not standalone or series based)

TP — Correctly found a relationship.
FP — Incorrectly predicted a relationship.
TN — Correctly rejected a non-relationship.
FN — Missed a real relationship.
Precision — Of predicted relationships, how many were correct.
Recall — Of real relationships, how many were found.
F1 — Balance between precision and recall.
FNR — Percentage of real relationships that were missed.
Specificity — Percentage of non-relationships correctly rejected.

Cosine similarity *compares the LLM-generated related-work text (the prequel/sequel/alt-version title+description) with all 8,108 book embeddings to find the closest existing node,* ***Cosine similarity ranges from -1 to 1*** *score of* ≥0.5 *is treated as a similar match, while* ***<0.5*** *is treated as dissimilar*

Coherence *is a post-matching sanity check, it compares the* ***central book's real, pre-computed corpus embedding*** *with the* ***matched book's real corpus embedding*** *from the 8,108-book dataset to see whether the two books are semantically related. It was added because exact-title matching could incorrectly match books from different series; about* ***47% of***

naive exact matches *in the 30-book test were such collisions, which coherence can flag when the two books have low semantic similarity.*

Coherence is most useful for ***exact-title matches****, where title agreement and embedding similarity provide independent evidence. For* ***cosine-selected matches****, it adds less information because cosine similarity already selected the most similar candidate;*

When an exact title match is confirmed to be from the ***same series****, we keep the match even if coherence is low because the series information is stronger evidence. When the series is unknown, coherence is used as an additional check for suspicious matches.*

Model	TP	FP	TN	FN	Precision	Recall	F1	FNR	Specificity
Llama 3.1 8B	108	17	148	57	0.864	0.655	0.745	0.345	0.897
Mistral 7B	107	58	107	58	0.648	0.648	0.648	0.352	0.648

Ambiguous set (n=70, no ground truth, informational only): Llama said True for 57/70,

6.3 Results — Full corpus, n=8,108 (3,665 positive / 4,443 negative)

Model	TP	FP	TN	FN	Precision	Recall	F1	FNR	Specificity
Llama (A)	3082	2051	2392	583	0.600	0.841	0.701	0.159	0.538
Mistral (A)	2309	2040	2403	1356	0.531	0.630	0.576	0.370	0.541
Llama (B)	3023	1558	2885	642	0.660	0.825	0.733	0.175	0.649
Mistral (B)	2057	1696	2747	1608	0.548	0.561	0.555	0.439	0.618

6.5 Matching stage, n=400

Model	Exact accepted	Exact rejected (series mismatch)	Cosine ≥ 0.5 (incl. reranked)	Cosine < 0.5	Flagged low-coherence

Llama 3.1 8B	34 (30 confirmed-series, 4 unverifiable)	30	75 (23 reranked)	43	81
Mistral 7B	37 (25 confirmed-series, 12 unverifiable)	32	98 (21 reranked)	59	103
Llama 3.1 8B (Version B)	**47**	**23**	**102**	**—**	**78**
Mistral 7B (Version B)	**36**	**27**	**94**	**—**	**94**

Matching stage, n=8,108

Model	Exact accepted	Exact rejected (series mismatch)	Cosine ≥ 0.5 (incl. reranked)	Cosine < 0.5	Flagged low-coherence
Llama 3.1 8B (Version A)	972	779	2,158	1,224	2,364
Mistral 7B (Version A)	737	573	1,739	1,300	1,790
Llama 3.1 8B (Version B)	1,054	622	2,061	843	1,939
Mistral 7B (Version B)	750	497	1,535	971	1,511

6.6 Observations

Llama gave fewer wrong relationship predictions than Mistral, with about **3.4× fewer false positives** while recall was similar.

Both models can get the **correct yes/no answer** but still give the **wrong book title or author**. **Precision only measures relationship correctness.**

The series-position reranking fixed some known cases, such as “Grab Your Pole” #2 → #1. However, it can miss cases where the same series appears under **different names** in the dataset, such as **“Cirque Du Freak”** and **“The Saga of Darren Shan.”**

7. Relation Type: SEQUEL

A sequel is a book that comes after the central book in the same series or universe. The prompt structure is the same as the Prequel prompt, but the direction is reversed.

Model	TP	FP	TN	FN	Precision	Recall	F1
Llama 3.1 8B	7	4	11	8	0.636	0.467	0.538
Mistral 7B	14	10	5	1	0.583	0.933	0.718

7.3 Results — Version A, n=400 (165 with_series/ 165 without_series / 70 ambiguous)

Model	TP	FP	TN	FN	Precision	Recall	F1	FNR	Specificity
Llama 3.1 8B	63	14	151	102	0.818	0.382	0.521	0.618	0.915
Mistral 7B	138	122	43	27	0.531	0.836	0.649	0.164	0.261

Ambiguous set (n=70, informational): Llama True for 15/70, False for 55/70; Mistral True for 53/70, False for 17/70.

7.3a Results — Full corpus, n=8,108 (3,665 positive / 4,443 negative)

Model	TP	FP	TN	FN	Precision	Recall	F1	FNR	Specificity
Llama (A)	3120	3168	1275	545	0.496	0.851	0.627	0.149	0.287
Mistral (A)	3218	3455	988	447	0.482	0.878	0.623	0.122	0.222
Llama (B)	3279	3297	1146	386	0.499	0.895	0.640	0.105	0.258
Mistral (B)	3216	3428	1015	449	0.484	0.877	0.624	0.122	0.228

7.4 Matching stage, n=400

Model	Exact accepted	Exact rejected	Cosine ≥ 0.5	Flagged
Llama 3.1 8B	22	21	36	49
Mistral 7B	30	54	155	158
Llama 3.1 8B (Version B)	**20**	**23**	**47**	**52**
Mistral 7B (Version B)	**37**	**49**	**146**	**135**

Matching stage, n=8,108

Model	Exact accepted	Exact rejected	Cosine ≥ 0.5	Flagged
Llama 3.1 8B (Version A)	742	878	3,146	3,045
Mistral 7B (Version A)	698	970	3,023	3,149
Llama 3.1 8B (Version B)	865	929	3,661	3,407

Mistral 7B (Version B)	892	936	3,190	3,094

Version B note: for Mistral, cosine similarity did most of the matching work in both versions (155 in A, 146 in B) this tell how many books Mistral predicted as sequels, not a change in cosine matching itself.

7.5 Observations

The same difference between the two models showed up again for Sequel,

1. **Mistral:** Recall = **0.836** → identified most of the actual Sequels, but also produced many false positives.
2. **Llama:** Recall = **0.382** → identified fewer of the actual Sequels, with fewer false positives.
3. This shows that Mistral was more likely to say a book was a Sequel, while Llama was more selective.
4. We also saw some differences between runs even though the code stayed the same. Since the temperature was set to **0.2**, the model can still produce different outputs from one run to another, so some variation is observed.

8. Relation Type: ALTERNATIVE VERSION

8.1 Definition

An Alternative Version is a materially different version of an existing story, characters, or core narrative premise. This includes alternate timelines, alternate universes or canons, explicit "what if" versions, and major adaptations into another medium such as graphic novels, manga, movies, TV, or anime.

8.3 Prompt evolution and a documented failure of a keyword-based guard

The **first prompt** gave too little guidance: **15/20 “yes,” but only 1/15 was defensible.**
It mainly failed because it lacked clear examples of what **should not** count,

Rewritten prompt: **3/20 “yes,” with the two previous hallucinations fixed, but one Sequel was still incorrectly marked as an Alternative Version.**

Code Guard v1: **Forced False when “sequel” or “prequel” appeared in reason, but failed because the model put the information in match_description instead.**

Code Guard v2: **Checked both fields, but the model bypassed the guard by rephrasing the relationship; this happened in 2 consecutive runs, so we stopped adding keyword-based rules,** kept this as a known limitation of the classification step and that can be handled by LLM classification → cosine/coherence matching some wrong predictions can still be filtered during the matching stage but the Alternative Version false-positive issue is still unresolved at the classification stage.

8.4 Results — Version A, n=30 (15 with_series/ 15 without_series)

Model	TP	FP	TN	FN	Precision	Recall	F1	FNR	Specificity
Llama 3.1 8B	6	1	14	9	0.857	0.400	0.545	0.600	0.933
Mistral 7B	4	0	15	11	1.000	0.267	0.421	0.733	1.000

The experiment was stopped at n=30 by explicit decision. Going beyond this would require manually researching and verifying each additional real-world adaptation, unlike Prequel and Sequel where the ground truth can be generated from structural signals. Given the time required and the results already available for the other two relation types, I decided not to scale it further.

8.5 Matching stage, n=30

Model	Claimed matches	Non-book medium (unmatched, no target corpus)	Exact accepted	Cosine ≥ 0.5	Flagged
Llama 3.1 8B	7	1	2	1	0
Mistral 7B	4	2	0	1	1
Llama 3.1 8B (Version B)	**4**	**0**	**1**	**1**	**1**
Mistral 7B (Version B)	**5**	**1**	**0**	**1**	**3**

similarity matched only 1 book per model in both versions, mainly because both models made very few Alternative Version predictions, not because the matching itself performed poorly..

8.6 Observations

Alternative Version had **high precision but low recall** (0.857 / 1.000; 0.400 / 0.267). The models were usually right when they said “yes,” but **missed most real adaptations** in sample Mistral also had **zero false positives and perfect specificity**, unlike its Prequel/Sequel behavior, and the reason for this difference is still unknown and should be treated as an open finding.

9. Cross-Relation-Type Model Behavior Summary

Relation	Model	Precision	Recall	Specificity	Note
Prequel (n=400)	Llama	0.864	0.655	0.897	conservative
Prequel (n=400)	Mistral	0.648	0.648	0.648	liberal
Sequel (n=400)	Llama	0.818	0.382	0.915	conservative, lower recall than Prequel
Sequel (n=400)	Mistral	0.531	0.836	0.261	most liberal result of the session
Alt. Version (n=30)	Llama	0.857	0.400	0.933	conservative
Alt. Version (n=30)	Mistral	1.000	0.267	1.000	reversed - most conservative result of the session

Across Prequel and Sequel, **Mistral produced more false positives and had lower specificity**, but this reversed for Alternative Version. One possible reason is that the **“genuinely rare” framing and stronger exclusions** list in Alternative Version prompt made Mistral more cautious, but this has not been tested.

12. Version A Results Summary

Relation	Model	Precision	Recall	F1	FNR	Specificity
Prequel (n=400)	Llama 3.1 8B	0.864	0.655	0.745	0.345	0.897
Prequel (n=400)	Mistral 7B	0.648	0.648	0.648	0.352	0.648
Sequel (n=400)	Llama 3.1 8B	0.818	0.382	0.521	0.618	0.915
Sequel (n=400)	Mistral 7B	0.531	0.836	0.649	0.164	0.261
Alt. Version (n=30)	Llama 3.1 8B	0.857	0.400	0.545	0.600	0.933
Alt. Version (n=30)	Mistral 7B	1.000	0.267	0.421	0.733	1.000
Prequel (n=8,108)	Llama 3.1 8B	0.600	0.841	0.701	0.159	0.538
Prequel (n=8,108)	Mistral 7B	0.531	0.630	0.576	0.370	0.541
Sequel (n=8,108)	Llama 3.1 8B	0.496	0.851	0.627	0.149	0.287
Sequel (n=8,108)	Mistral 7B	0.482	0.878	0.623	0.122	0.222

13. Version B Results - All Three Relation Types

Relation	Model	Precision (A→B)	Recall (A→B)	F1 (A→B)	Specificity (A→B)	Net effect
Prequel (n=400)	Llama 3.1 8B	0.864 → 0.894	0.655 → 0.764	0.745 → 0.824	0.897 → 0.909	improved on every axis
Prequel (n=400)	Mistral 7B	0.648 → 0.705	0.648 → 0.624	0.648 → 0.662	0.648 → 0.739	improved (precision/ specificity up, recall flat)

Sequel (n=400)	Llama 3.1 8B	0.818 → 0.796	0.382 → 0.448	0.521 → 0.574	0.915 → 0.885	improved (F1 up, small precision/specificity trade-off)
Sequel (n=400)	Mistral 7B	0.531 → 0.569	0.836 → 0.848	0.649 → 0.681	0.261 → 0.358	improved on every axis
Alt. Version (n=30)	Llama 3.1 8B	0.857 → 1.000	0.400 → 0.267	0.545 → 0.421	0.933 → 1.000	regressed (fewer true positives found: 6→4)
Alt. Version (n=30)	Mistral 7B	1.000 → 0.600	0.267 → 0.200	0.421 → 0.300	1.000 → 0.867	regressed (fewer true positives found: 4→3)
Prequel (n=8,108)	Llama 3.1 8B	0.600 → 0.660	0.841 → 0.825	0.701 → 0.733	0.538 → 0.649	improved
Prequel (n=8,108)	Mistral 7B	0.531 → 0.548	0.630 → 0.561	0.576 → 0.555	0.541 → 0.618	regressed (recall down)
Sequel (n=8,108)	Llama 3.1 8B	0.496 → 0.499	0.851 → 0.895	0.627 → 0.640	0.287 → 0.258	improved
Sequel (n=8,108)	Mistral 7B	0.482 → 0.484	0.878 → 0.877	0.623 → 0.624	0.222 → 0.228	almost no change

Adding the author to the embeddings improved F1 for both models on both Prequel and Sequel. All 4 comparisons improved. For Alternative Version, F1 decreased for both models. So, adding the author may not improve every relation type but need to verified against rest of 6 relationship types we have

On the full corpus (n=8,108), adding the author improved F1 for Llama on both relations, made almost no change for Sequel Mistral, and lowered F1 for Prequel Mistral, so the n=400 result did not hold for Mistral. Llama with Version B had the best F1 for both relations (Prequel 0.733, Sequel 0.640).

Verified precision, n=8,108

Verified precision = correct / (correct + confirmed incorrect). Unknown predictions are reported separately. Correct: the book has the relation in the ground truth. Confirmed incorrect: the ground truth has no relation and the title named by the LLM is in the 8,108 books. Unknown: the

ground truth has no relation and the named title is not in the 8,108 books, so it may be a real book missing from the dataset. Correct is checked at relationship level, not title level.

Relation (Version)	Model	Correct	Confirmed incorrect	Unknown	Verified precision	Precision
Prequel (A)	Llama 3.1 8B	3,082	513	1,538	0.857	0.600
Prequel (A)	Mistral 7B	2,309	448	1,592	0.838	0.531
Prequel (B)	Llama 3.1 8B	3,023	431	1,127	0.875	0.660
Prequel (B)	Mistral 7B	2,057	408	1,288	0.834	0.548
Sequel (A)	Llama 3.1 8B	3,120	670	2,498	0.823	0.496
Sequel (A)	Mistral 7B	3,218	698	2,757	0.822	0.482
Sequel (B)	Llama 3.1 8B	3,279	727	2,570	0.819	0.499
Sequel (B)	Mistral 7B	3,216	760	2,668	0.809	0.484

Mac vs. CCR, same 330 labelled books

The n=400 runs were on the Mac and the n=8,108 runs were on CCR. With the same prompts and temperature, the two gave the same answer for 44.8% to 80.6% of the same books per run. For example, Sequel Llama Version B recall was 0.448 on the Mac and 0.915 on CCR. So the two sets of numbers should not be read as sample vs. population. The cause has not been tested.

13a. Execution Timing Summary

Relation	Version	Model	Books	Total time	Avg/call
Prequel	A	Llama 3.1 8B	400	~80 min	~12.00s
Prequel	A	Mistral	400	~95 min	~14.25s

Prequel	B	Llama 3.1 8B	400	72.8 min	10.91s
Prequel	B	Mistral	400	80.6 min	12.09s
Sequel	A (flawed ground truth, n=30)	Llama 3.1 8B	30	15 minutes	30.00s
Sequel	A (flawed ground truth, n=30)	Mistral	30	20 minute	40.00s
Sequel	A (corrected ground truth, n=30)	Llama 3.1 8B	30	15.3 min	30.60s
Sequel	A (corrected ground truth, n=30)	Mistral	30	16.1 min	32.20s
Sequel	A	Llama 3.1 8B	400	90.6 min	13.59s
Sequel	A	Mistral	400	83.1 min	12.47s
Sequel	B	Llama 3.1 8B	400	80.7 min	6.85s
Sequel	B	Mistral	400	70.2 min	8.28s
Alt. Version	A (final frozen), n=30	Llama 3.1 8B	30	13.2 min	26.40s
Alt. Version	A (final frozen), n=30	Mistral	30	14.9 min	29.80s
Alt. Version	B, n=30	Llama 3.1 8B	30	13.2 min	26.40s
Alt. Version	B, n=30	Mistral	30	10.4 min	20.80s
Prequel	A (CCR)	Llama 3.1 8B	8,108	306 min	8.16s
Prequel	A (CCR)	Mistral	8,108	220 min	5.99s
Prequel	B (CCR)	Llama 3.1 8B	8,108	243 min	6.53s
Prequel	B (CCR)	Mistral	8,108	218 min	5.99s

Sequel	A (CCR)	Llama 3.1 8B	8,108	164 min	4.47s
Sequel	A (CCR)	Mistral	8,108	273 min	7.07s
Sequel	B (CCR)	Llama 3.1 8B	8,108	168 min	4.58s
Sequel	B (CCR)	Mistral	8,108	269 min	6.96s

Mistral was consistently slower than Llama in every comparable run, the difference ranged from about 1.1× to 1.5×, averaging around 1.3×, the LLM extraction work took approximately 12.9 hours in total, spread across two overnight sessions. This does not include matching/scoring, which took less than a minute per run, or the 12.7-minute Version B embedding regeneration.

The CCR times are Slurm job times with 4 concurrent requests per job, about 31 hours in total, so they are not comparable with the Mac times.

Prequel:

You are a book-knowledge assistant. You are given the title of a book. Using your own knowledge of this book's universe, its series, and its fictional universe, determine whether it has a PREQUEL — a book set chronologically or narratively before it, in the same series or universe.

Book title: "{title}"

Think through the evidence first, then give your answer. Respond with ONLY a JSON object, no other text, in exactly this format:

```
{
  "series_info": "series name and position if known, or empty string if unknown",
  "reason": "step-by-step: does this book's series/universe have an earlier book? name it if so, or state clearly why not",
  "confidence": "high" or "medium" or "low",
  "has_prequel": true or false,
  "prequel_title": "exact title of the prequel, or empty string if none",
  "prequel_author": "author name, or empty string if none",
  "prequel_description": "1-2 sentence plot summary of the prequel, or empty string if none"
}
```

If you are not confident a prequel exists, or the book is standalone, return has_prequel: false with empty strings for the other fields. Do not invent a prequel that does not exist.

Sequel:

You are a book-knowledge assistant. You are given the title of a book. Using your own knowledge of this book's universe, its series, and its fictional universe, determine whether it has a SEQUEL — a book set chronologically or narratively AFTER it, in the same series or universe.

Book title: "{title}"

Think through the evidence first, then give your answer. Respond with ONLY a JSON object, no other text, in exactly this format:

```
{
  "series_info": "series name and position if known, or empty string if unknown",
  "reason": "step-by-step: does this book's series/universe have a later book? name it if so, or state clearly why not",
  "confidence": "high" or "medium" or "low",
  "has_sequel": true or false,
  "match_title": "exact title of the sequel, or empty string if none",
  "match_author": "author name, or empty string if none",
  "match_description": "1-2 sentence plot summary of the sequel, or empty string if none"
}
```

If you are not confident a sequel exists, or the book is standalone/the final entry, return has_sequel: false with empty strings for the other fields. Do not invent a sequel that does not exist.

Alternative Version:

You are a book-knowledge assistant. You are given the title of a book. Using your own knowledge, determine whether this book has an ALTERNATIVE VERSION.

DEFINITION: Alternative Version means a materially different narrative version of an existing story, characters, or core narrative premise. This includes:
- An alternate timeline where a major event has a different outcome.
- An alternate-universe version using the same characters or core premise.
- An explicit "what if" version of the original story.
- A substantial graphic novel or manga adaptation of the same novel.
- A movie, TV, or anime adaptation that presents the same underlying narrative as a distinct work.
- A deliberately altered alternate-canon version of the original story.

The relationship MUST come from the underlying narrative being changed, branched, or substantially adapted. The following do NOT qualify, even if they seem related:
- A normal sequel or prequel.
- A companion novel in the same continuous canon.
- A spin-off or side story.

- A simple jump forward or backward in time (that's still the same canon, not an alternative version).
- A different cover or ordinary illustrated edition.
- An abridged/unabridged edition without narrative changes.
- A book that only references or shares characters with another work.
- Same title belonging to an unrelated author or story (a title collision is NOT evidence of a relationship).

Book title: "{title}"

Think through the evidence first, then give your answer. Respond with ONLY a JSON object, no other text, in exactly this format:
{
  "series_info": "series name and position if known, or empty string if unknown",
  "reason": "step-by-step: is there a materially different narrative version (alt-timeline/alt-canon/adaptation)? If you name one, explain specifically HOW the narrative differs, not just that a related book/adaptation exists. If none, state why the candidates you considered don't qualify (e.g. 'X is a companion novel in the same canon, not a narrative branch').",
  "confidence": "high" or "medium" or "low",
  "has_alternative_version": true or false,
  "match_title": "exact title of the alternative version, or empty string if none",
  "match_author": "author/studio/creator name — leave this EMPTY if you are not certain, do not guess an author for a title you're unsure about",
  "match_description": "1-2 sentences on specifically how the narrative diverges, or empty string if none",
  "match_medium": "book, movie, tv, or anime — whichever the alternative version actually is, or empty string if has_alternative_version is false"
}

Preq_ Category	Details
Total items	400
Standalone	112 (28.0%)
Series-numbered	288 (72.0%)
Ground truth (288 numbered)	148 positive · 83 negative · 57 other/unclear
Positive without number	17 positive-stratum books identified via r_sequel

Series position (n=288)	Min 0.3 · Median 2.0 · Mean 3.76 · Max 96.0
Distinct series	274
1 item in sample	261
2 items	12
3+ items	1
Most represented	Sweet Valley High 3 items

Sequel_l Category	Details
Total items	400
Standalone	53 (13.2%)
Series-numbered	347 (86.8%)
Ground truth (347 numbered)	151 positive · 165 negative · 31 other/unclear
Standalone negatives	0 — negative rule requires confirmed series position
Series position (n=347)	Min 0.3 · Median 2.0 · Mean 3.85 · Max 76.0
Distinct series	319
1 item in sample	295

2 items	20
3+ items	4
Most represented	DC Icons 3 items

14. What These Results Demonstrate, and What They Do Not

Demonstrated LLMs find relationships: They can identify Prequels, Sequels, and Alternative Versions that HSAL may miss. Models differ: Llama and Mistral make different predictions, so they cannot be treated as interchangeable. Titles can mislead: Books with the same title may belong to different series, so series information helps avoid false matches. Self-correction is unreliable: Asking the LLM to check its answer again does not consistently fix missed relationships. Keyword rules fail: Simple rules based on words like "sequel" can miss relationships when the LLM uses different wording.

Not yet demonstrated HSAL graph integration: Recall@K/NDCG@K impact not tested yet. Version B vs. A: Improves Prequel/Sequel at n=400; on the full corpus improves Llama, mixed for Mistral; decrease Alternative Version. Mac vs. CCR difference: cause not tested. Top-K sensitivity: K=1, 3, 5 not tested yet. Cross-dataset matching: Anime/MovieLens matching not attempted yet. Remaining five categories: No definitions or pipelines yet.

15. Immediate Next Steps

Get the remaining definitions: get definitions of the other 5 taxonomy categories before expanding the project further. Start HSAL integration: Read HSAL_for_BPR1.py and KGReader.py to understand the graph structure, then integrate Prequel and Sequel relationships using Version B embeddings, since they performed better. Defer other datasets: Anime and MovieLens embedding experiments will be done later, not in the current scope.